



DevOps Engineer course syllabus



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Təlimçi: Elşən Fərzəliyev

DevOps engineer

Təcrübə:

- Dövlət Xəzinədarlığı Agentliyi - DevOps komanda rəhbəri
- İKTex MMC - DevOps Engineer
- AccessBank Azerbaijan - DevOps Engineer
- CHİ Software - DevOps Engineer

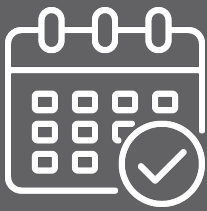


Fəaliyyətlər:

- Bakı Ali Neft Məktəbi - Linux/DevOps müəllimi
- IDTech Academy - Direktor
- ITalks Azerbaijan - Həm təsisçi
- Topdone.az - Həm təsisçi/CTO

Sertifikatlar: RHCSA, AWS Solution Architect, CKA, ITILv4 Foundation, Prince2, MCSA, MTA

Kursun formatı:



4 ay
32 dər
Həftədə 2 dəfə



Hər dər 2 saat davam
edir



Hər dərsdən sonra
şagirdlərə ev tapşırığı
verilir

Tədris Məqsədləri və Kursun Proqramı

Modul 1

Introduction to DevOps

- Monolithic architecture
- Microservice architecture
- Introduction to cloud
- Introduction to Container Orchestration
- Introduction to devops & CI/CD
- Devops tools
- Rolling updates demo on LMS

Modul 2

Code version controlling systems

- Git Overview
- Installing Git
- Git File Management, git remote & git local
- Core git concepts, branch, merge, rebase, cherry pick, stash, pull, push, tags
- Advanced GIT concepts, git hooks
- Git branching best practices, GitFlow with SDLC
- Git Practice

Modul 3

Docker

- Benefits of Containers
- Docker Overview
- Images & Containers
- Docker volumes
- Docker network
- Docker File
- Entrypoint, CMD & RUN
- Docker in Docker
- Docker Compose
- LAB: Running LMS microservices in local with Docker Compose

Modul 4

Configuration Management – Ansible

- Introduction to configuration management
- Infrastructure as Code (IaC) with tools like Ansible
- Managing infrastructure and application configuration
- Automating server provisioning
- Maintaining consistency across environments

Modul 5

KV Store - HashiCorp Vault

- Introduction to Vault
- Vault concepts
- Provisioning Vault Cluster

Modul 6

Kubernetes

- Kubernetes Foundation
- Namespaces
- Resource quota for Namespaces.
- Tricks and Tips with command
- Labels Selectors
- Kubernetes
- Deployments, Services, Replicas
- K8STemplates

Modul 7

Helm

- Helm Overview
- Using Helm Charts
- Custom Helm Charts
- Advanced Helm Templating operations
- Howto effectively use helm and k8s namespaces to support multiple envs (Stage/Dev)
- LAB: Packaging and Deploying LMS microservices on Kubernetes with helm charts

Modul 8

Kubernetes Networking

- Network Overview
- Pod to Pod, Container to Container communication
- Kubernetes Services
- MetalLB
- Ingress
- External Traffic into Kubernetes
- LAB: Expose LMS Microservices through Ingress

Modul 9

Limits & Configurations

- Config Maps
- Secrets
- Mount ConfigMap and Secret
- Kubernetes Probes(Liveness and readiness checks)
- Resource limits
- LAB: Multiple labs on GCP servers in the cloud

Modul 10

Managing the cluster

- Taint and Toleration
- Node Affinity
- Node Selector
- Metric Server
- Applications Logs
- Command and Arguments
- Init Containers
- OS Upgrade
- Upgrade Cluster
- Backup and Restore methods
- LAB: Rolling updates and Roll back in deployments

Modul 11

Managing Roles & State

- Role and Role Bindings
- ClusterRole and Cluster Rolebindings
- EmptyDir, HostPath
- Storage Types
- Persistent volume and Persistent Volume Claims
- Cron Jobs
- StateFul Sets
- RBAC
- LAB: Persisting data for LMS microservices

Modul 12

ELK Stack

- Introduction to elastic stack
- Collecting microservice logs in EFK
- LAB: Create dashboard and monitor the microservices

Modul 13

Prometheus

- Introduction to Prometheus
- Setup & Configure
- Prometheus Query Language
- Monitoring
- Metrics & alerts
- LAB: Create dashboard in Grafana and monitor the microservices

Modul 14

CI/CD

- CI/CD tools, Gitlab, etc.
- Gitlab CI overview
- Deep dive on the LMS CI/CD implementation
- Gitlab runners
- Automate the whole SDLC process E2E
- LAB: Create a fully functional CI and CD pipeline for LMS Microservices in Gitlab

Modul 15

GitOps

- GitOps Tools
- Introduction to ArgoCD
- Deploy with ArgoCD Application
- Managing repository and application
- Manual and Auto sync of Deployments
- LAB: Deploy MS with ArgoCD

Modul 16

IaaS – Terraform

- Managing terraform
- Creating Cloud infrastructure using Terraform
- Main terraform configurations

Modul 17

DevOps in Practice and Final Projects:

- Real-world case studies and success stories
- Group or individual projects applying DevOps practices
- Project presentations and demonstrations
- Recap of key concepts and practices learned